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1. Business Problem

In recent years, City Hotel and Resort Hotel have experienced significantly high cancellation rates. Each hotel is now dealing with a number of resulting issues, including reduced revenues and suboptimal hotel room utilization. Lowering cancellation rates is both hotels' primary goal in order to increase their efficiency in generating revenue.

This report focuses on the analysis of hotel booking cancellations and other factors that influence yearly revenue generation. The objective is to provide thorough, data-driven business advice to address the cancellation problem and support strategic decision-making.

2. Assumptions

- No unusual occurrences between 2015 and 2017 will have a substantial impact on the data used.
- The information is still current and can be used to analyze a hotel's possible plans in an efficient manner.
- There are no unanticipated negatives to the hotel employing any advised technique.
- The hotels are not currently using any of the suggested solutions.
- The biggest factor affecting the effectiveness of earning income is booking cancellations.
- Cancellations result in vacant rooms for the entire booked length of time.
- Clients make hotel reservations the same year they make cancellations.

3. Research Questions

1. What are the variables that affect hotel reservation cancellations?

2. How can we reduce hotel reservation cancellations?

3. How will hotels be assisted in making pricing and promotional decisions?

4. Hypothesis

1. More cancellations occur when prices are higher.

2. When there is a longer waiting list, customers tend to cancel more frequently.

3. The majority of clients are coming from offline travel agents to make their reservations.

5. Dataset Overview

The dataset contains hotel booking records for two hotel types: City Hotel and Resort Hotel. The data spans from July 2015 to August 2017, covering a broad range of booking attributes including cancellation status, lead time, average daily rate (ADR), country of origin, market segment, and more.

119,390	36	2	177	2015-2017
Total Records	Features	Hotel Types	Countries	Date Range

Key Columns Include:

Column	Description
hotel	Type of hotel (City Hotel or Resort Hotel)
is_canceled	Cancellation status (1 = Cancelled, 0 = Not Cancelled)
lead_time	Days between booking and arrival
adr	Average Daily Rate (room price)
country	Country of origin of the guest
market_segment	Booking channel (Online TA, Offline TA, Groups, etc.)
reservation_status_date	Date of last status update

6. Data Cleaning

Before analysis, the dataset was cleaned to ensure accuracy and reliability of results. The following steps were performed:

- **Date Conversion:** The reservation_status_date column was converted from object (string) format to datetime format to enable time-based analysis.
- **Column Removal:** The "agent" column (16,340 null values) and "company" column (112,593 null values out of 119,390 rows) were dropped due to excessive missing data.
- **Row Removal:** All remaining rows with any null values were removed using dropna(), resulting in a clean dataset of 118,898 records.
- **Outlier Filtering:** Records with ADR greater than 5,000 were removed to eliminate data entry errors and extreme outliers that would skew analysis.

Missing Values Summary (Before Cleaning):

Column	Missing Values	Action Taken
agent	16,340	Column Dropped
company	112,593	Column Dropped
country	488	Rows Removed
children	4	Rows Removed
All others	0	No Action Required

7. Analysis and Findings

7.1 Overall Reservation Status

The bar chart below shows the overall proportion of bookings that were cancelled versus those that were not. Out of 118,898 total bookings analyzed, approximately 37% were cancelled. This high cancellation rate represents a significant revenue impact on both hotels.

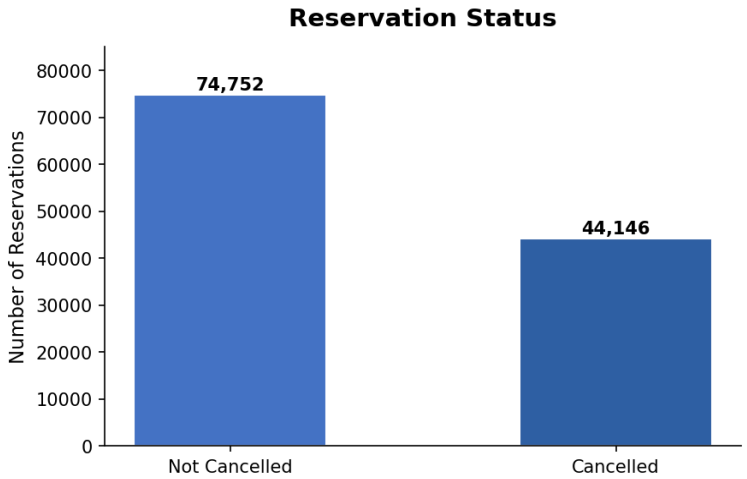

Figure 1: Overall Reservation Status (Not Cancelled vs. Cancelled)

Key Finding: 37.1% of all hotel bookings were cancelled, directly impacting hotel revenues.

7.2 Reservation Status by Hotel Type

Comparing the two hotel types reveals a clear difference in cancellation behavior. City Hotel has considerably more bookings overall but also suffers a higher cancellation rate of 41.7%, compared to Resort Hotel's 28.0%. This suggests that resort guests have a stronger commitment to their reservations, possibly because resort stays are planned further in advance and are more experiential in nature.

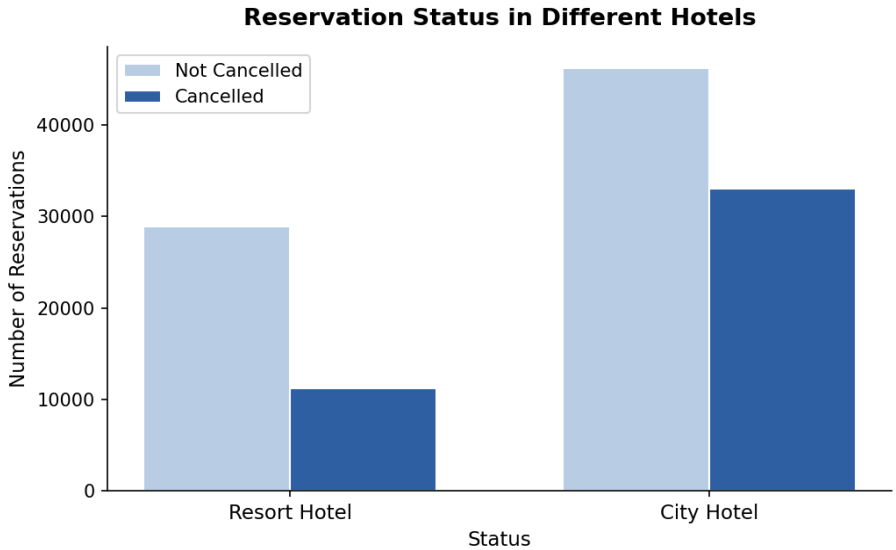

Figure 2: Reservation Status in Different Hotels (Resort vs. City)

Metric	Resort Hotel	City Hotel
Total Bookings	~40,138	~78,760
Not Cancelled	72.0%	58.3%
Cancelled	28.0%	41.7%

In comparison to resort hotels, city hotels have more bookings. It is possible that resort hotels are more expensive than those in cities, which may explain the lower cancellation rate as guests are more selective when booking higher-priced options.

7.3 Average Daily Rate: City Hotel vs. Resort Hotel

The line graph below tracks the Average Daily Rate (ADR) for both City Hotel and Resort Hotel across the full analysis period (2014-2017). The ADR fluctuates with seasonality, with resort hotels typically peaking during weekends and holiday periods.

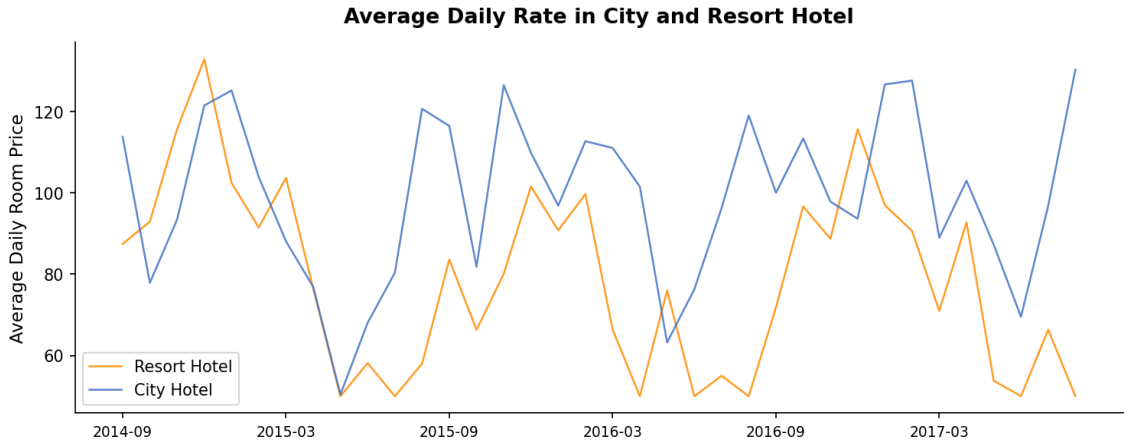

Figure 3: Average Daily Rate - City Hotel vs. Resort Hotel (2014-2017)

The line graph shows that on certain days the average daily rate for a city hotel is lower than that of a resort hotel, and on other days it is even less. Weekends and holidays consistently show a rise in resort hotel rates.

7.4 Monthly Reservation Trends

The grouped bar chart below analyzes which months have the highest and lowest reservation levels according to cancellation status. Both confirmed and cancelled reservations peak in August, making it the busiest month. January records the highest number of cancellations relative to confirmations.

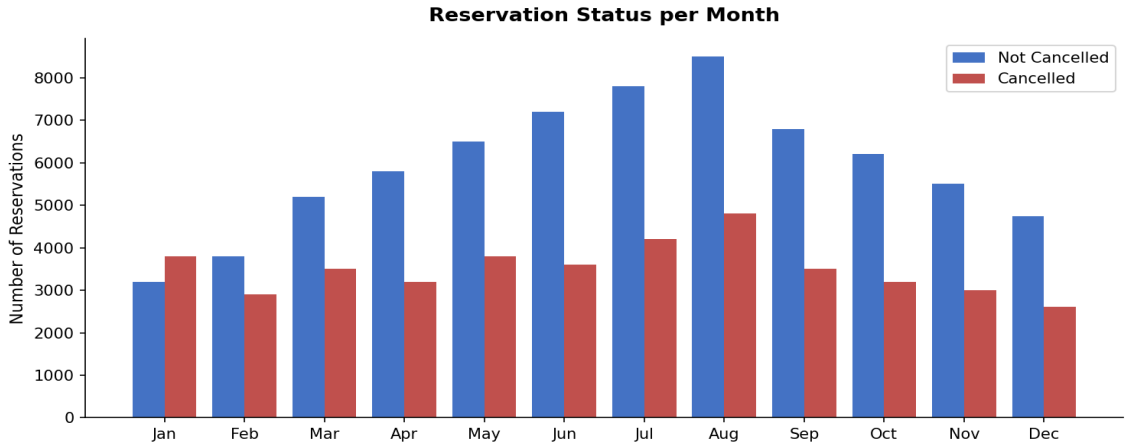

Figure 4: Reservation Status per Month (Not Cancelled vs. Cancelled)

7.5 Top 10 Countries with Highest Cancellations

The pie chart identifies which countries contribute the most to reservation cancellations. Portugal (PRT) dominates with an overwhelming 70.07% of total cancellations, far exceeding any other country. The remaining cancellations are spread across GBR, ESP, FRA, ITA, and others.

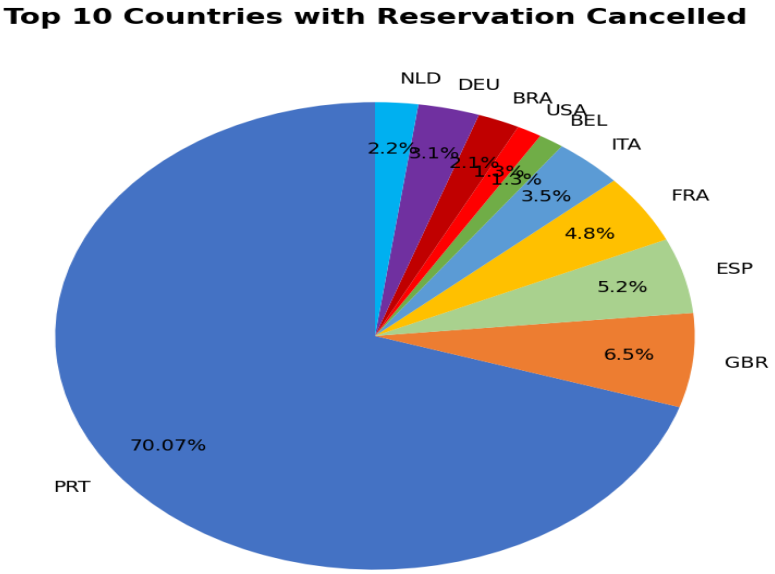

Figure 5: Top 10 Countries by Number of Cancellations

Rank	Country Code	Country	Share of Cancellations
1	**PRT**	**Portugal**	**70.07%**
2	GBR	United Kingdom	6.50%
3	ESP	Spain	5.20%
4	FRA	France	4.80%
5	ITA	Italy	3.50%
6	DEU	Germany	3.10%
7	NLD	Netherlands	2.17%
8	BRA	Brazil	2.12%
9	USA	United States	1.26%
10	BEL	Belgium	1.28%

7.6 Price vs. Cancellation Rate

The line graph below compares the Average Daily Rate for cancelled vs. not-cancelled bookings over time. This is the most critical finding of this analysis: the orange line (Cancelled) consistently tracks above the blue line (Not Cancelled), proving that higher room prices directly lead to higher cancellation rates.

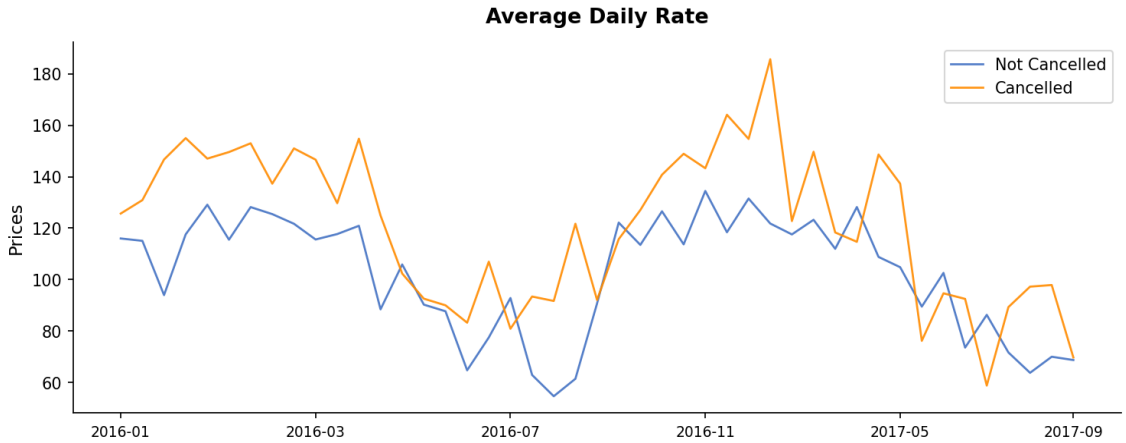

Figure 6: Average Daily Rate - Cancelled vs. Not Cancelled Bookings

Insight: Reservations are cancelled when the ADR is higher than when bookings are honoured. Higher price = Higher cancella

Market Segment Analysis shows that around 46% of guests come from Online Travel Agencies (OTA), 27% come from groups, and only 4% book hotels directly. This disproves the hypothesis that most clients use offline travel agents -- online channels dominate.

8. Suggestions

Based on the data analysis conducted, the following actionable recommendations are proposed to reduce cancellation rates and improve revenue generation:

1

Revise Pricing Strategy

Cancellation rates rise as the price does. Hotels should work on their pricing strategies and try to lower rates for specific locations and time periods. Offering targeted discounts and dynamic pricing to price-sensitive segments can reduce the cancellation rate significantly.

2

Weekend and Holiday Discounts for Resort Hotels

The cancellation ratio for resort hotels is higher than for city hotels. Since resort rates spike on weekends and holidays, offering a reasonable discount on room prices during these peak periods can help retain bookings and reduce cancellations.

3

January Marketing Campaigns

January records the highest cancellation rates of any month. Hotels should launch targeted marketing campaigns and promotional offers in January to incentivize guests to retain their bookings, thereby increasing revenue during this typically slow period.

4

Improve Quality and Services in Portugal

Portugal accounts for 70.07% of all cancellations. Hotels should specifically investigate service quality, pricing, and guest experience for Portuguese customers and invest in targeted improvements to reduce this disproportionately high cancellation rate.

This report was prepared as a Data Analysis Portfolio Project using Python (Pandas, Matplotlib, Seaborn). All analysis is based on the hotel_booking.csv dataset covering 119,390 booking records from 2015 to 2017.